Confidence-Aware Demand and Inventory Planning Tool Project Proposal

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Abstract

Retail forecasting is an imprecise practice that is prone to error, which can cause significant profit losses. Fixed safety stock buffers based on singular conventional forecasts, along with naively opaque AI approaches to demand estimation, can be overconfident and magnify the impacts of incorrect forecasts. To bring transparency to the process of retail demand estimation, we propose to build a proof-of-concept AI-based retail forecasting dashboard that visualizes actual sales, imputed demand for missing-not-at-random data during stockout events, and 7-day forecasts using the HuggingFace-hosted FreshRetailNet-50k dataset [1]. We will first evaluate several neural forecasting models and uncertainty quantification method combinations with respect to demand estimation error (measured with weighted absolute prediction error for magnitude and weighted prediction error for bias direction) as well as uncertainty estimation calibration (measured by smoothed expected calibration error) and sharpness (measured with mean prediction interval width). Models are considered successful if they achieve error within 1 standard deviation of prior published results [2] and are properly calibrated. After the assessment phase, we will deploy the best combination through a REST API and dashboard for downstream interaction by demand planners and inventory stockers.

1 Introduction and Problem Statement

Retailers must decide how much of each product to stock, and errors cost both ways: too little causes stockouts and lost sales, while too much ties up capital and leads to markdowns or spoilage. Analysts estimate this "inventory distortion" at roughly $1.7 trillion a year, about 6.5% of global retail sales [3], though methods vary. Planners turn a single forecast per product into safety stock and reorder points, but that number does not say how reliable it is, so they must accept it unchecked or override it by judgment, which cannot be sustained across thousands of products. The problem is likely worst for new and niche products, which have little history and tend to get the least review. During a stockout, recorded sales fall to zero, so later forecasts may read unmet demand as weak demand and cut stock further, while aggregate accuracy metrics can still look healthy. *Direct stakeholders*, who interact with the system or its output, are demand planners and inventory managers, who set stock levels, and customers, who experience the output through shelf availability. *Indirect stakeholders*, who are affected without interacting with it, are suppliers, warehouse staff, and finance, who absorb forecast errors as order swings, emergency restocks, and excess stock. The most exposed are customers and suppliers of new and niche products (full map in Appendix A).

The non-AI alternative is rule-based planning: a fixed safety-stock buffer on a single forecast, with manual overrides. It is transparent but gives no product-specific signal of reliability, and assuming normal forecast error likely fits poorly for intermittent, promotion-driven, stockout-censored demand, a hypothesis we will test. Probabilistic forecasting, such as quantile regression and conformal prediction, gives each product a prediction interval. Width expresses *uncertainty*; *calibration* is the separate property that nominal 80% intervals contain realized demand about 80% of the time. A wide interval from an uncalibrated model is not a reliable warning, so the system needs both. A confidence label derived from interval width then routes high-uncertainty products to human review.

Selbst et al. (2019) [4] caution that the best solution may not involve technology, and planner time is limited, which no model can create. We argue this is not solutionism because the system only makes unreliable forecasts visible and sets the order of review, while planners keep the stocking decision. Because routing is a value judgment about whose products get attention, we will examine how review is allocated across high- and low-volume products.

2 Literature Review and Baseline

Most machine learning work on demand forecasting gives one number. Douaioui et al. [5] reviewed 119 studies (2015–2024) on machine learning (ML) and deep learning (DL) for supply-chain demand forecasting. Abbasimehr et al. [6] showed that an LSTM beat classic statistical methods, such as ARIMA and exponential smoothing, and other ML methods on a furniture company's demand. Mahajan and Pattanshetti [7] fed weather and social-media and news mood into combined DL models, and used SHAP technique to show which inputs contributed most to particular predictions and in which direction. The Temporal Fusion Transformer (TFT) proposed by Lim et al. [8] can predict a low, middle, and high value at once and shows which inputs mattered. Chronos [9], [10] gives range forecasts out of the box and did about as well as, and sometimes better than, models trained on each new dataset, but it does not yet use extra inputs such as the days a product is on sale. For our project, this shows that ML can beat simple methods and use signals

like weather. But most studies judge a model only by how close one number is to real sales, so a planner still cannot tell a safe forecast from a risky one.

Other work gives forecasts with a range. Makridakis et al. [11] built the Walmart sales forecasts on 42,840 series as nine quantiles instead of one number. It shows that ranges for retail sales can be forecast and scored at scale. Conformal prediction is a method that wraps around any trained model and gives a range, or a short set of answers, that holds the true value at a rate the user picks, such as 90% of the time [12]. Bruys et al. [13] used it on DL models at a large parcel company, and the answer sets hit their promised rates (99% and 95%) on test data. Gibbs and Candès [14] adapted it to keep that rate even when the data shifts over time. Closest to our work was done by Zhao et al. [15], a 2026 preprint used conformal ranges to plan inventory on FreshRetailNet-50K and cut cost by about 8–27% compared with three other methods. For our project, this means honest ranges can be added to almost any model and kept honest when demand changes, which links to our "demand has changed" warning.

Two more problems shape our data: stockouts hide demand, and some products are hard to forecast. Wang et al. [2] released FreshRetailNet-50K, our dataset: hourly sales for 50,000 store-product series (863 fresh products, 898 stores, 18 cities) with hourly out-of-stock flags. When an item sells out, sales show less than people wanted. Filling in this hidden demand lowered a TFT forecaster's error from 31.75% to 29.02% and cut a 7.37% under-forecast to almost zero. Error stayed highest for low-selling products, at about 37%. For our project, the stockout flags let us avoid learning from falsely low sales, and the four types tell us which products will be hard to forecast and will likely get the widest ranges.

Recent work also warns that a better forecast score does not always mean better stock decisions. Using M5 data, Theodorou et al. [16] found that when a lost sale costs more than holding extra stock, the most accurate method is often not the best one for inventory, especially for products that sell irregularly. For our project, this suggests looking at stockouts and waste, not only at forecast error.

Taken together, we found no work that combines honest ranges, a warning when demand changes, and sending only uncertain products to a planner, on fresh-retail data where stockouts hide real demand. We also found no peer-reviewed study that checks whether the ranges from pre-trained models are honest on retail demand, and the closest work on our dataset is still a preprint [15]. Our project aims to fill this gap.

3 System Architecture and Methodology

3.1 Data Pipeline

We use the preprocessed FreshRetailNet-50k [1] (CC BY 4.0 [2]) for a proof-of-concept forecasting dashboard. Following [2], we first recover latent demand from missing-not-at-random (MNAR) stockout data, then forecast from real and imputed data. With hourly sales $\mathbf{y}$ and in-stock indicator $s_{j,k,t} \in \{0, 1\}$ for item j, store k, hour t:

$$\mathbf{d} = \mathbf{y} \odot \mathbf{s} + \hat{\mathbf{d}} \odot (1 - \mathbf{s}) \tag{1}$$

$$\hat{\mathbf{d}} = f_\theta(\mathbf{y} \odot \mathbf{s}, \mathbf{s}, \text{covariates}) \tag{2}$$

Daily demand $\mathbf{D}_T = \sum_{t \in T} \mathbf{d}_t$ is then forecast seven days ahead:

$$\hat{\mathbf{D}}_{T+1:T+7} = f_\phi(\mathbf{D}_{T-H:T}, \text{covariates}_{T-H:T+7}) \tag{3}$$

The goal is trustworthy weekly product-need estimates per store. If the dataset becomes unavailable, we have three fallbacks: Kaggle Favorita [17] (closest, but not hourly and lacking hierarchy), M5 [18], and Walmart Recruiting [19] (smaller and coarser).

3.2 Modeling Framework

Models. Following [2], [20], we will implement two imputation–forecasting pairs: DLinear–DLinear (most interpretable) and TimesNet–Temporal Fusion Transformer (TFT) (expected to provide strong imputation and forecasting performance, respectively). We will compare their WAPE to evaluate whether their forecasting performance differs significantly in practice. DLinear [21] splits a series into a moving-average trend and a remainder, applies a linear layer to each, and sums them, giving fully linear, interpretable forecasts at low cost. TimesNet [22] uses the FFT to find k dominant periods, reshapes the series into 2D tensors, and applies Inception convolutions [23] to capture intra- and inter-period patterns. It had the best WAPE, WPE, and $\rho_{\text{demand-supply}}$ but is less interpretable. TFT [8] combines variable selection networks, a static covariate encoder, LSTM encoder/decoder, gated residual networks, and masked multi-head attention to output quantile forecasts. We will evaluate TFT as a forecasting model and compare its performance and interpretability with DLinear. Architecture diagrams for the proposed models will be included in Appendix E (figures 2, 3, 4).

Uncertainty quantification. For each pair, we will evaluate post-hoc methods: conformal prediction [12], test-time data augmentation [24], [25], [26], and an external uncertainty network [27]. Conformal prediction will score calibration points using a nonconformity score $s(x, y)$, take the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile $\hat{q}$, and return $\mathcal{C}(x_{\text{test}}) = \{y \mid s(x_{\text{test}}, y) \leq \hat{q}\}$, guaranteeing $\mathbb{P}(y_{\text{test}} \in \mathcal{C}) \geq 1 - \alpha$; the main difficulty will be choosing a score function for each model. Test-time augmentation will aggregate predictions over perturbed inputs, but since existing work targets image classification, we will need to design augmentations that introduce sufficient randomness without pushing inputs off-distribution. The external network will be a feedforward model trained using the interval loss of [27].

3.3 System Architecture

The system is a batch pipeline (laptop for tests, Northeastern's Explorer cluster [28] for full runs). The diagram is in Appendix E (figure 1). We build an hourly series per store–product pair and mark stockout hours. A recovery model fills those hours (Eqs. 1, 2), hours are summed to daily totals, and a forecaster predicts seven days (Eq. 3), always as a matched pair. Conformal prediction wraps each forecast in a range; its width sets a confidence label, and a "demand has changed" warning fires when recent sales repeatedly fall outside their ranges, as in adaptive conformal inference [14]. Every pair and method is scored on validation splits (Section 4), and only the best combination feeds the dashboard via one shared table (store, product, day) with a run log of data, code, and settings (Appendix C). A small web API serves three views: a review queue of uncertain or changed products, a product explanation page, and a model-health page with coverage and fairness by sales group. Planner overrides are logged with reasons and signal when retraining is needed. The system stops at forecasts and ranges and does not connect to live ordering.

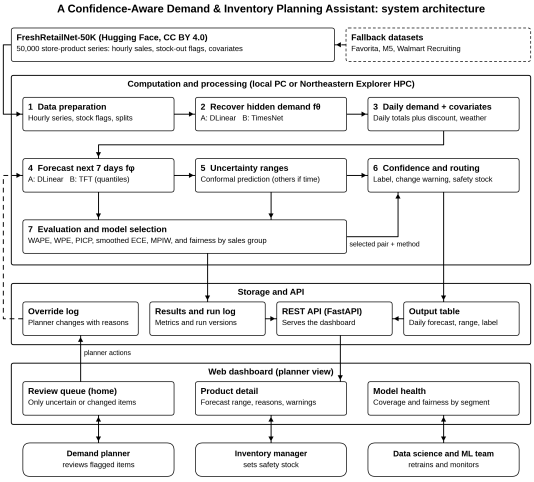


Figure 1: Anticipated Dashboard/Model System Architecture

3.4 The Human Who Acts

Demand planners and inventory managers use each forecast and range to set stock levels. We design first for when *the system is wrong* [29]: each product shows its range, confidence label, and change warning; uncertain products are never applied automatically; every override is logged with a reason. We expect *overtrust* [30] to be the larger risk: if intervals under-cover, a risky product could look "confident," skip review, and stock out, so we test coverage overall and separately for low- and high-selling products. *Undertrust* is secondary: planners may ignore wide ranges on new or niche products, tying to our fairness question.

3.5 Workplan

Each of four members leads one stream (data, dashboard, evaluation, compute), implements one base model and one uncertainty method, and writes one report section. The output format is fixed in week 7 so streams run in parallel; conformal prediction is the priority method, with the others as stretch goals (Appendix D).

4 Evaluation Metrics and Validation

4.1 Metrics

The system must be evaluated on two dimensions: the accuracy of the imputations and forecasts, and the calibration of the uncertainty estimates behind them. For accuracy, FreshRetailNet-50k [1], [2] provides standard demand-forecasting metrics. Weighted Absolute Percentage Error (WAPE, 4) measures the magnitude of forecast error for forecast F versus actual demand A, and Weighted Percentage Error (WPE, 5) drops the absolute value to measure the direction of bias.

$$WAPE = 100\% \cdot \frac{\sum_{t=1}^{n} |A_t - F_t|}{\sum_{t=1}^{n} A_t} \tag{4}$$

$$WPE = 100\% \cdot \frac{\sum_{t=1}^{n} (A_t - F_t)}{\sum_{t=1}^{n} A_t} \tag{5}$$

Because stockouts censor demand, no ground truth exists for those periods, so imputation metrics are evaluated only on non-censored data, and the forecasting task uses the imputed values to fill censored periods. Although the original paper does not evaluate it, we may also assess how accurately stockout periods can be classified, to test whether stockouts can be anticipated.

For calibration, a 90% confidence interval should contain the true value 90% of the time. Most calibration methods target classification [31], [32], but [33] extends them to regression by treating a forecaster as outputting a CDF F_t for each timestep. Calibration is typically assessed with expected calibration error (ECE) and reliability diagrams [31], [32], [33], both based on empirical coverage, which for regression is

$$\hat{p}_l = \frac{|\{y_t | F_t(y_t) \leq p_l \quad \forall t \in T\}|}{T} \tag{6}$$

for confidence level p_l (related to prediction interval coverage percentage, PICP [34]). A discrete ECE over L confidence levels, weighted by the number of points n_l in each band, is

$$\text{dECE} = \frac{1}{N} \sum_{l=1}^{L} n_l \cdot |p_l - \hat{p}_l|. \tag{7}$$

To avoid the problems of discretization, we plan to use a kernel-density-based smoothed ECE [35], which to our knowledge would be its first application to regression.

ECE alone cannot separate multiple estimators alone, since several may achieve similar coverage with different sharpness. We would therefore also report mean prediction interval width (MPIW) [34], and jointly assess smoothed ECE (overall calibration), reliability diagrams (calibration by confidence band), and MPIW (sharpness).

4.2 Validation

FreshRetailNet-50k [2] ships with a fixed train/test split, which we will use only for the final evaluation. During model and UQ comparison, we will create our own 80/20 train/validation splits from the training data and evaluate each model-UQ combination on at least 3 random splits, to check that performance differences are practically and statistically significant.

5 Ethical Considerations, Deployment, and Risk

To check for bias, we use two items from the AI fairness checklist of Madaio et al. [36]: *2.3:Define and scrutinize fairness criteria* and *3.3: Assess fairness criteria*. We apply them to one named group: **suppliers of new or niche (low-selling) products**. These products have little sales history, so they get wide ranges, are flagged "low confidence" more often, and may wait longest in the planner's queue. In the dataset paper [2], forecast error is about 37% for low-selling items but 23–28% for high-selling items. Our fairness rule is that we report error, under-forecasting, and how often the range holds the real sales separately for low- and high-selling products. Low-selling products must not be under-forecast more often and must not always sit at the back of the review queue. If the prototype breaks this rule, we fix it before calling it done.

On privacy, the dataset we use holds no personal data. It has hourly sales and stock status for each store-product pair, plus discount, holiday and activity flags, and weather. Stores, cities, and products appear only as number codes, and there are no customer records. Our purpose needs only these fields, and we will not add outside data that could reveal a store or supplier [37]. If a store asked for its records to be deleted, we would drop every series with that store's code, re-run data cleaning, retrain the model, recompute the ranges, and re-run evaluation. Because the codes are anonymous, the data publisher would have to tell us which code to remove.

To keep the output clear, the planner sees for each product the forecast, its range, a confidence label, any "demand has changed" warning, the main reasons (for example a promotion, rain, a holiday, or a recent trend), how often past ranges held the real sales, and whether past sales were cut short by stockouts. This follows the guideline **make clear how well the system can do what it can do** [29]. Uncertain or changed products are never applied automatically, and a person decides. Large jumps in safety stock need review. New and niche products get a reserved review slot and a minimum stock level, and every planner override is logged with a reason.

The system will be deployed as a research prototype. The team runs it as a batch job on the public dataset, and it does not connect to any live ERP or ordering system. The team maintains it during the semester. In a real pilot, the ML engineer (who retrains and monitors the model) and the ERP administrator (who owns the data feed) from our stakeholder map would take over. The next iteration starts if change warnings fire on many products at once, ranges miss real sales much more often than promised, the error gap between low and high selling products grows, planners override most flagged products, or new data arrives (more seasons, real stock levels, or lead times). No written scoping conditions were set for our project, so we keep the limits from the project brief: public data only, no live ERP connection, a research prototype, and a bounded set of products and demand patterns.

The seed ethical question our team will carry through the semester's ethics check-ins is: *When the system sends uncertain products to a planner, whose products wait at the back of the line? Is it fair that new, small, or niche products, which have the least data, keep getting marked "unreliable" and end up with too little stock?*

Appendices

A: Team Stakeholder Map

Table 1: Team stakeholder map

Stakeholder	**Category**	**Gains if it works**	**Loses if it is wrong (error named)**	**In the data?**
Demand planner who reviews flagged products	Direct	Reviews only the products the confidence label and change warning flag, not the full catalog	**Missed flag:** a risky product is labeled confident, skips review, and stocks out. **False flag:** safe products are flagged, wasting time and eroding trust in the tool.	**No.** No planner decisions or review logs.Past stock decisions are not attributed.
Inventory manager who sets safety stock and reorder points	Direct	Safety stock and reorder points sized from each product's forecast interval, not one number	**Under-forecast:** stockouts. **Over-forecast:** excess stock, markdowns, or spoilage. **Miscalibrated interval:** buffers sized from a range that misstates real variability.	**Partly.** Hourly stockout flags show past failures. No on-hand stock, safety stock, reorder points, or lead times.
Customer buying the product	Indirect	Fewer stockouts on the products they want, including slower-selling ones	**Under-forecast:** the item is out of stock and the unmet demand leaves no record. **Over-forecast:** excess or older stock.	**Partly.** Aggregate hourly sales only, with no customer records. Stockout demand appears as zeros.
Warehouse and store operations staff	Indirect	Fewer emergency restocks, because stock is sized to forecast uncertainty	**Under-forecast:** emergency restocks and overtime. **Over-forecast:** unsold stock to handle or discard.	**Partly.** Store and group IDs and stockout flags. No restock events, staffing, or costs.
Suppliers of established products	Indirect	More stable order volumes, because orders follow calibrated forecasts	**Over-forecast:** cancelled orders from unsold stock. **Under-forecast:** smaller orders. **False change alert:** sudden swings in orders.	**No.** No supplier IDs, purchase orders, or lead times. Products are anonymized.
Suppliers and customers of new and niche products	Indirect	A fair chance to show demand: an honest wide range instead of an unexamined low forecast	**Wide interval, low-confidence label:** the product waits in the review queue and is stocked cautiously, stocks out, and the recorded zeros lower the next forecast.	**Partly.** Product and category IDs and sparse series identify low-volume products. There is no launch-date field, so "new" cannot be confirmed.
Head of Supply Chain or executive management who approves the pilot	Gatekeeper	Evidence of when forecasts can and cannot be trusted before approving a rollout	**Missed flag on a high-impact product:** a stockout or waste spike they cannot explain. **Miscalibrated intervals:** they approve the rollout on overstated confidence.	**No.** Service-level targets and waste budgets are not in the data.

Purchasing or store manager who approves orders	Gatekeeper	Order sizes backed by a stated range and confidence label	**Wrongly confident forecast:** approves a large order that proves wrong. **Under-forecast:** approves a short order and runs out.	**No.** No order quantities, minimum order sizes, or lead times.
Finance lead who funds the project and approves order budgets	Gatekeeper	Less capital tied up in stock, if intervals avoid blanket over-buffering	**Over-forecast across many products:** more money in stock than promised, so the project is cut. **Overly wide intervals:** blanket extra buffers tie up capital.	**No.** Only a discount rate with no prices, costs, or margins, so the financial case cannot be tested.
Data science and ML team that retrains and monitors the model	Maintainer	Change warnings and interval-coverage checks that show when retraining is needed	**Missed demand change:** the model is quietly wrong for weeks. **Coverage drift:** intervals stop matching real variability unnoticed.	**No.** The observation window is short, so long-run drift and seasonality cannot be tested.
Software, ERP, and infrastructure engineers	Maintainer	A stable interface and a clean data connection	**Late or broken data feed:** forecasts built on stale data are shown with unchanged confidence labels.	**No.** A cleaned research extract. Behavior of a live feed is unknown.

Note: This map reflects team analysis and the FreshRetailNet-50K documentation. because collectors never use the output, and the fixed public dataset holds no price or cost data for profit analysis. Tarakaram alone named the ERP administrator, and Jeena alone listed sparse-history products as their own row. We kept both, folding them into the engineers row and the new-and-niche row.

Jeena combined the demand planner and inventory manager, while Tarakaram separated them, and we kept them separate. Tarakaram's system suggests safety stock and reorder points, while Gil's and Jeena's stop at forecasts. We followed the latter, so people set stock levels from the ranges. Jeena listed the store manager as direct, while Tarakaram treated purchasing managers as gatekeepers. We chose gatekeeper because they approve orders rather than set stock levels. For the overlooked stakeholder, Tarakaram chose suppliers of new and niche products, Gil customers, and Jeena shoppers with unrecorded demand. We kept new and niche suppliers as our fairness group and kept unrecorded demand in the customer row. All three maps listed customers as indirect, so we list them as indirect. We dropped Gil's dynamic-pricing concern because Appendix C rules out pricing decisions.

B: Datasheet

Primary dataset: FreshRetailNet-50K [2]. The questions follow Gebru et al. [38]. "Unknown" means that neither the dataset paper nor the Hugging Face dataset card answers the question.

Question	Answer
Motivation	
For what purpose was the dataset created?	To give researchers a public benchmark for demand that is hidden when a product sells out. According to the dataset paper, other retail datasets (M5, Favorita, Walmart) do not mark stockouts and report sales only daily or weekly, while this one marks stockouts hour by hour.
Who created the dataset, and on behalf of which entity?	Eight researchers: six at Dingdong Ltd., a fresh e-commerce company in China, one at Sun Yat-Sen University, and one unaffiliated. The data come from Dingdong's own operations, and the dataset is released under Dingdong's Hugging Face account.
Who funded its creation?	**Unknown.** No funder is named.
Any other comments?	Released under CC-BY-4.0. The dataset card links the paper and a GitHub repository with the training and evaluation pipeline.
Composition	
What do the instances represent?	One product in one store on one day: daily and hourly sales, hourly stock status, and that day's discount, holiday, and weather values. Each row is a day within one of 50,000 store-product time series, about 4.85 million rows in total.
How many instances are there?	About 4.85 million daily rows (4.5 million train and 350 thousand evaluation), forming 50,000 store-product series from 898 stores in 18 Chinese cities. Each series has 90 training days, and the evaluation split adds about 7 days per series. The paper reports 863 perishable SKUs and the dataset card reports 865. In total this is about 0.1 billion hourly observations.
All possible instances, or a sample?	**Unknown.** The paper and dataset card do not say whether the 50,000 series are complete or a selection. They average about 56 products per store (50,000 / 898) out of 863 SKUs, and cover one retailer over about three months, so they are probably a subset. The selection rule is also **unknown**.
What data does each instance consist of?	Each record has anonymized IDs (city, store, management group, three category levels, and product), the date, and daily sales. It also has 24 hourly sales values, 24 hourly out-of-stock flags (1 means out of stock), and a count of out-of-stock hours between 6:00 and 22:00. Contextual fields are the discount rate, holiday flag, activity flag, precipitation, average temperature, humidity, and wind level. Sales values are globally normalized.
Is there a label or target?	No single label field is designated. The paper defines two tasks: recovering the demand hidden during stockouts and forecasting daily demand 7 days ahead. Observed sales (daily and hourly) are the values models learn from and are scored against. The hourly out-of-stock flags are annotations that show where sales are censored. True demand during stockouts is not provided, so the paper tests recovery by simulating censoring on periods without stockouts.
Is any information missing?	Yes. There are no prices (only a discount rate), on-hand quantities, supplier or order data, customer records, or product names (IDs only). Sales are globally normalized by an undisclosed coefficient, so absolute units are unknown. Demand during stockouts is unobserved by design.
Are relationships between instances explicit?	Yes. Stores nest in cities (city, then store). Products nest in categories (management group, three category levels, then product). Time order is given by date and hour.

Question	Answer
Are there recommended data splits?	Yes. Hugging Face provides a train split (4.5 million rows) and an evaluation split (350 thousand rows). This is about 90 training days and 7 evaluation days per series. The paper's forecasting task has a 7-day horizon.
Are there errors, noise, or redundancies?	Stockout labels are described as verified, but the verification method is **unknown**. Sales are sparse and zero-inflated, and the paper reports higher forecast error for low-selling items (WAPE about 37%, versus 23–28% for high-selling items). Weather values are daily, not hourly. Redundancy: **unknown**.
Is it self-contained?	Yes, apart from the original weather and holiday sources. Weather came from an external API and holidays from the Chinese statutory calendar, but the values are included in the data. Long-term availability guarantees: **unknown**.
Does it contain confidential data?	It holds commercially sensitive store-level sales, released by the owner with encrypted store IDs. No personal confidential data.
Might any content be offensive or distressing?	No. The fields are numeric records and anonymized IDs.
Does it identify subpopulations?	Not people. The data are organized by city, store, and product category. The paper also compares high- and low-selling items, but that is an analysis grouping and not a field. There are no demographics.
Can individuals be identified?	Not from the released fields: sales are aggregates and there are no customer fields. Whether encrypted store IDs could be re-identified through outside data: **unknown**.
Is any data sensitive?	No personal, financial, or health data about people, and no store locations (only encrypted IDs).
Any other comments?	The data come from one retailer in China, so patterns may not transfer to other markets or to non-perishable goods.
Collection Process	
How was the data acquired?	From Dingdong's own systems: stock levels from warehouse management systems, sales from online order transactions, discount rates from marketing campaign records, daily weather from an API, and holiday labels for Chinese statutory holidays. Sales and stock are directly observed. Stockout flags come from hourly stock records, the exact derivation is **unknown**.
What mechanisms or procedures were used?	Warehouse management and order-transaction systems, plus a weather API. How the mechanisms were validated: **unknown**.
What was the sampling strategy?	**Unknown.** The paper does not say how the 50,000 series were chosen.
Who collected the data, and how were they compensated?	**Unknown.** The paper does not describe the collectors. Most authors are Dingdong employees.
Over what timeframe was it collected?	A three-month observation period, March to June 2024, according to the paper. In the series viewed on Hugging Face, the training data run from 28 March to 25 June 2024. The evaluation dates are **unknown**. The paper was first posted in May 2025.
Was an ethical review conducted?	**Unknown.** None is mentioned.
Was it collected from individuals or via third parties?	Not collected from individuals for research. Sales aggregate customer orders on Dingdong's platform, and weather came from a third-party API.
Were individuals notified?	**Unknown.**
Did they consent?	**Unknown.**
Can consent be revoked?	**Unknown.**
Was an impact analysis on data subjects conducted?	**Unknown.** The paper says personal information was pseudonymized and store IDs encrypted before release, but reports no formal impact assessment.
Any other comments?	This datasheet is based on the dataset paper and the Hugging Face dataset card. We did not contact the authors.

C: Governance Plan

This plan follows the four elements of data governance from Verhulst et al., [39]: Why, How, Who, and What. It covers our primary dataset, FreshRetailNet-50K.

Why: the purpose, and the uses we rule out

We use the data to build and test a research prototype that gives fresh-product forecasts with honest ranges and change warnings, and to check that it treats low-selling products fairly.

Uses we rule out:

- Judging or ranking stores, warehouses, or their staff.
- Trying to find out which real stores, cities, products, or suppliers the number codes stand for.
- Making live ordering or pricing decisions.

How: our principles

- **Use only what the purpose needs**. Decision: we keep only the fields listed in Section 5 and do not join any outside data that could identify a store or supplier.
- **Report every group, not just the average**. Decision: no result counts as final until it is shown separately for low-selling and high-selling products.
- **Keep every result traceable.**Decision: each model run records the data version, the code version, and the settings used, so anyone can repeat it.

Who: the people responsible

- Access is limited to team members and course staff.
- Who is accountable: the whole team is accountable for how we use the data and for the prototype's outputs. The data publisher [2] is accountable for how the data was first collected.

What: practices across the data's life

- We collect no new data. We download FreshRetailNet-50K from its public Hugging Face page (Dingdong-Inc/FreshRetailNet-50K) and record the download date and version.
- The data is kept in a private team folder. Raw data is never pushed to a public code repository.
- We check data quality, keep the test period separate from training, and report results by product sales group.
- We keep the data and models until the course ends. Then we delete our raw copies and keep only code and summary results. Anyone can download the data again from the source.

D: Work Plan

Week	Date	Course milestone	Tarakaram	Jeena	Gil	Vishnu
5	Oct 13	The Craft of Storyboarding	Download data, record version, set up repo	Dashboard storyboard	Dashboard storyboard	Environment, compute plan
6	Oct 20	Interaction Walkthrough	Build series, mask stockout hours	Dashboard walkthrough	WAPE and WPE code	Test neuralforecast on a sample
7	Oct 27	Architecture and Tooling Seminar; Midpoint Design Review	Time-based splits, DLinear imputation	DLinear forecasting	PICP and MPIW code, start TimesNet	Start TFT, fix shared output format
8	Nov 3	Pipeline skeleton due	End-to-end pipeline run	Dashboard skeleton on shared format	Smoothed ECE, reliability diagrams	Both pairs run end to end
9	Nov 10	Ethics and impact milestone due	Conformal prediction	Test-time augmentation	Fairness report by sales group. Ethics write-up	External interval network
10	Nov 17	Deployment & Monitoring	Conformal results across splits	Confidence label, change warning	External parametric network	Monitoring and retraining plan
11	Nov 24	Cybersecurity for AI Systems **(Minimum Working System)**	Method comparison	Reasons and review queue	Results tables, by group	Remaining runs. Data security check
12	Dec 1	Continual Learning & Dress Rehearsal	Datasheet draft	Dashboard polish, demo	Governance plan, final evaluation	Final runs on official eval split
13	Dec 8	Final Showcase	Present data	Present dashboard	Present evaluation	Present models
14	Dec 15	Final report and impact assessment due	Data section	Interface section	Evaluation, ethics section	Models section

Table 3: Work plan by week and team member.

E: Diagrams and Figures

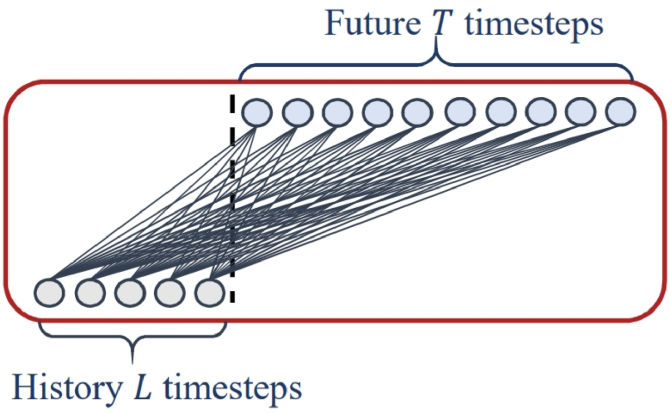


Figure 2: DLinear architecture diagram [21]

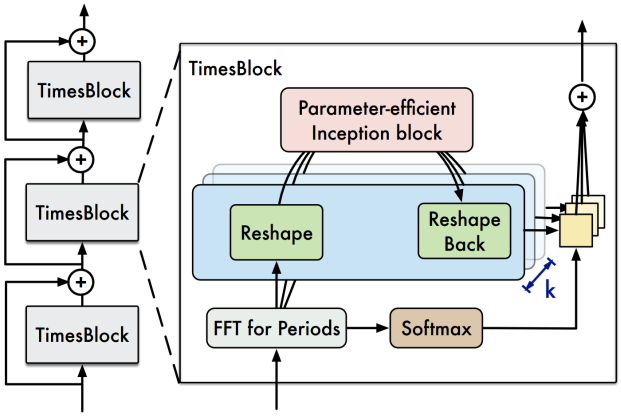


Figure 3: TimesNet architecture diagram [22]

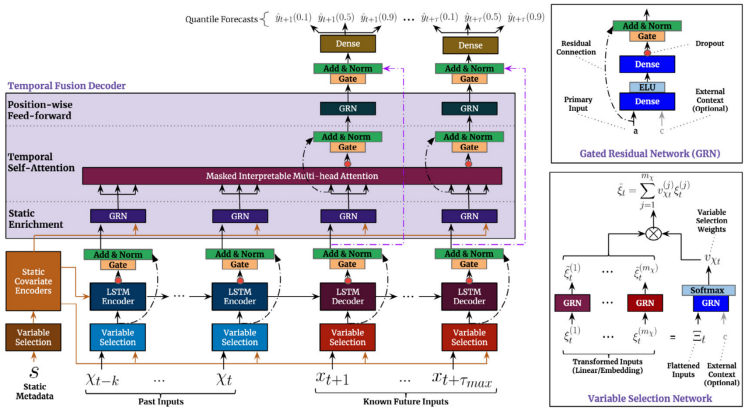


Figure 4: Temporal Fusion Transformer architecture diagram [8]

F: AI Usage Disclosure Statement

We disclose that the system architecture diagram was drafted with AI assistance, and there was editing assistance with respect to the text in the document for clarity and succinctness to fit within page limits. Everyone in the group contributed their own domain expertise and created content initially without AI-based assistance.

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